

```
import numpy as np

def cosine_similarity(a, b):

    return np.dot(a, b) / (np.linalg.norm(a) * np.linalg.norm(b))

good = np.array([0.82, 0.41, 0.10])
great = np.array([0.79, 0.38, 0.12])
bad = np.array([-0.71, 0.12, -0.05])

print(f"good and great are {cosine_similarity(good, great):.3f}")
print(f"good and bad are {cosine_similarity(good, bad):.3f}")
print(f"great and bad are {cosine_similarity(great, bad):.3f}")
```

```
good and great are 1.000
good and bad are -0.808
great and bad are -0.816
```

```
!pip install gensim
from gensim.models import Word2Vec

sentences = [
    'the king ruled the kingdom wisely',
    'the queen ruled the land gracefully',
    'the prince will become king one day',
    'a man walked into the palace',
    'a woman walked into the palace',
    'the dog barked loudly in the garden',
    'the puppy played happily in the garden',
]

tokenised = [s.split() for s in sentences]

model = Word2Vec(sentences=tokenised, vector_size=50,
window=3, min_count=1, sg=1, epochs=200)

print(model.wv.most_similar('king', topn=3))

result = model.wv.most_similar(
    positive=['king', 'woman'],
    negative=['man'],
    topn=1
)
print('king - man + woman =', result[0][1])
```

```
Requirement already satisfied: gensim in /usr/local/lib/python3.12/dist-packages (4.4.0)
Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.12/dist-packages (from gensim) (2.0.2)
Requirement already satisfied: scipy>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from gensim) (1.16.3)
Requirement already satisfied: smart_open>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from gensim) (7.6.1)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1->gensim) (2.2.1)
[('kingdom', 0.40141454339027405), ('played', 0.38476690649986267), ('in', 0.38395753502845764)]
king - man + woman = 0.2322036474943161
```

```
import sys
!{sys.executable} -m pip install gensim

!wget --no-check-certificate https://nlp.stanford.edu/data/glove.6B.zip
!unzip -q glove.6B.zip

from gensim.scripts.glove2word2vec import glove2word2vec
glove2word2vec('glove.6B.100d.txt', 'glove.6B.100d.w2v.txt')

from gensim.models import KeyedVectors
glove = KeyedVectors.load_word2vec_format(
    'glove.6B.100d.w2v.txt', binary=False
)
print(glove.most_similar('bollywood', topn=5))

res = glove.most_similar(positive=['paris','tamilnadu'], negative=['france'],
topn=1)
print('paris - france + india =', res[0][0])

print(glove.similarity('dog', 'puppy'))
print(glove.similarity('cat', 'cricket'))
```

```
Requirement already satisfied: gensim in /usr/local/lib/python3.12/dist-packages (4.4.0)
Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.12/dist-packages (from gensim) (2.0.2)
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--2026-06-11 05:52:28-- https://nlp.stanford.edu/data/glove.6B.zip
Resolving nlp.stanford.edu (nlp.stanford.edu)... 171.64.67.140
Connecting to nlp.stanford.edu (nlp.stanford.edu)|171.64.67.140|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://downloads.cs.stanford.edu/nlp/data/glove.6B.zip [following]
--2026-06-11 05:52:28-- https://downloads.cs.stanford.edu/nlp/data/glove.6B.zip
Resolving downloads.cs.stanford.edu (downloads.cs.stanford.edu)... 171.64.64.22
Connecting to downloads.cs.stanford.edu (downloads.cs.stanford.edu)|171.64.64.22|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 862182613 (822M) [application/zip]
Saving to: 'glove.6B.zip.2'

glove.6B.zip.2 16%[==>] 134.05M 5.04MB/s eta 2m 4s ^C
replace glove.6B.50d.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: /tmp/ipykernel_29924/3113185206.py:11: DeprecationWarning: Call to deprecated `glove2word2vec('glove.6B.100d.txt', 'glove.6B.100d.w2v.txt')`
[('malayalam', 0.678368151878967), ('film', 0.660624086856842), ('films', 0.6603192090988159), ('hindi', 0.6582645773887634), ('movie', 0.6446917
paris - france + india = bhubaneswar
0.7236376
0.1919626
```

```

import numpy as np
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA
from gensim.models import KeyedVectors
import os
from gensim.scripts.glove2word2vec import glove2word2vec

# Check if the GloVe word2vec format file exists, if not, create it
glove_w2v_path = 'glove.6B.100d.w2v.txt'
if not os.path.exists(glove_w2v_path):
    print("GloVe word2vec format file not found. Attempting to download and convert...")
    # These commands are from McbpMepgDCB4, ensuring dependencies are met
    !wget --no-check-certificate https://nlp.stanford.edu/data/glove.6B.zip
    !unzip -q glove.6B.zip
    glove2word2vec('glove.6B.100d.txt', glove_w2v_path)
    print("GloVe word2vec format file created.")
else:
    print("GloVe word2vec format file already exists.")

# Load GloVe model
glove = KeyedVectors.load_word2vec_format(
    glove_w2v_path,
    binary=False
)

# Words to visualize
words = [
    'king', 'queen', 'man', 'woman', 'prince', 'princess',
    'dog', 'puppy', 'cat', 'kitten', 'rama chandra', 'shravanth',
    'paris', 'france', 'delhi', 'india', 'london', 'england'
]

# Keep only words that exist in the model
valid_words = [w for w in words if w in glove.key_to_index]

missing_words = set(words) - set(valid_words)
if missing_words:
    print(f"Missing words from GloVe model: {missing_words}")

# Extract vectors
vectors = np.array([glove[w] for w in valid_words])

print("Vector matrix shape:", vectors.shape)

# Reduce dimensions from 100D -> 2D
pca = PCA(n_components=2, random_state=42)
coords = pca.fit_transform(vectors)

# Plot
fig, ax = plt.subplots(figsize=(10, 7))

ax.scatter(
    coords[:, 0],
    coords[:, 1],
    s=80,
    color='steelblue',
    zorder=3
)

# Add labels
for i, word in enumerate(valid_words):
    ax.annotate(
        word,
        (coords[i, 0], coords[i, 1]),
        fontsize=11,
        xytext=(5, 4),
        textcoords='offset points'
    )

ax.set_title(
    'GloVe Word Clusters (PCA Projection to 2D)',
    fontsize=13
)

ax.axhline(0, color='gray', linewidth=0.5)
ax.axvline(0, color='gray', linewidth=0.5)

plt.tight_layout()
plt.savefig('word_clusters.png', dpi=150)
plt.show()

```

GloVe word2vec format file not found. Attempting to download and convert...

--2026-06-11 08:04:24-- <https://nlp.stanford.edu/data/glove.6B.zip>

Resolving nlp.stanford.edu (nlp.stanford.edu)... 171.64.67.140

Connecting to nlp.stanford.edu (nlp.stanford.edu)|171.64.67.140|:443... connected.

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Location: <https://downloads.cs.stanford.edu/nlp/data/glove.6B.zip> [following]

--2026-06-11 08:04:24-- <https://downloads.cs.stanford.edu/nlp/data/glove.6B.zip>

Resolving downloads.cs.stanford.edu (downloads.cs.stanford.edu)... 171.64.64.22

Connecting to downloads.cs.stanford.edu (downloads.cs.stanford.edu)|171.64.64.22|:443... connected.

HTTP request sent, awaiting response... 200 OK

Length: 862182613 (822M) [application/zip]

Saving to: 'glove.6B.zip'

glove.6B.zip 100%[=====>] 822.24M 5.07MB/s in 2m 39s

2026-06-11 08:07:04 (5.16 MB/s) - 'glove.6B.zip' saved [862182613/862182613]

/tmp/ipykernel_8580/2209259068.py:15: DeprecationWarning: Call to deprecated `glove2word2vec` (KeyedVectors.load_word2vec_format(..., binary=False, glove2word2vec('glove.6B.100d.txt', glove_w2v_path)

GloVe word2vec format file created.

Missing words from GloVe model: {'rama chandra', 'shravanth'}

Vector matrix shape: (16, 100)

GloVe Word Clusters (PCA Projection to 2D)

